

Benchmarking a Physics-Informed Neural Network Against a Physics-Only Regression and Gaussian Process Regression for Reverse Osmosis Membrane Degradation Diagnosis

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Abstract—Physics-informed neural networks (PINNs) have been proposed for reverse osmosis membrane modeling and degradation diagnosis, via a data loss plus a physics loss weighted by λ . The original study evaluated this only against its own $\lambda = 0$ configuration, leaving open which component drives its performance. We benchmark the PINN, on the same dataset and protocol, against thirteen models (classical regressors, ensembles, a support vector machine, a multilayer perceptron, and Gaussian process regression, GPR) and a network-free regression of the same six transport parameters (“physics-only”). The physics loss, read literally as specified, has zero gradient with respect to the network weights; networks trained at $\lambda \in \{1, 10, 100\}$ are numerically identical to one at $\lambda = 0$, so “PINN vs. NN” is not meaningful, and no swept learning rate, width, or depth closes the gap with physics-only. Against this baseline ($R^2 = 0.918$ for permeate salinity), physics-only reaches $R^2 = 0.947$ and a 56% stronger anomaly ratio, reproducing the study’s defect-fraction (β) rise on degraded data; a paired bootstrap puts this advantage at $\Delta R^2 = +0.033$ (95% CI [0.000, 0.082]), a modest but consistent effect. GPR reaches similar accuracy ($R^2 = 0.935$) and a stronger anomaly ratio with no transport equations, but this does not survive bootstrap testing (95% CI [−0.011, 0.063]); its uncertainty is also flat across healthy and degraded conditions and misses the anomaly entirely. The PINN’s contribution is thus its interpretable output, not an established accuracy edge; physics-only matches that interpretability with a modest, bootstrap-supported advantage and no training instability.

Index Terms—Reverse osmosis, physics-informed neural networks, Gaussian process regression, model benchmarking, membrane degradation, anomaly detection

Highlights

- The physics loss, read literally, has zero gradient with respect to the network weights.
- PINN and the plain network are numerically identical for $\lambda \in \{0, 1, 10, 100\}$.
- A network-free physics-only regression is the best of 15 models on permeate salinity.

- Physics-only gives a bootstrap-supported $\Delta R^2 = +0.033$ and a 1.56× stronger anomaly signal.
- It reproduces the defect-fraction (β) rise on degraded data with no network and lower variance.
- Gaussian process predictive uncertainty is flat across conditions and misses the degradation.

I. INTRODUCTION

Reverse osmosis (RO) is a widely used water purification technology for municipal desalination, industrial water recycling, and brackish water treatment. Long-term operation degrades RO membranes through fouling, scaling, chemical attack, or mechanical stress, impairing water quality and system performance [1]. Operational strategies such as flow-reversal RO have been piloted to mitigate degradation and improve recovery [2], but robust modeling and diagnosis of degradation remain essential complements to such strategies. Machine learning (ML) has been applied extensively to RO performance modeling [3], [4], design optimization [5], [6], fouling and integrity monitoring [7]–[9], process control [10], [11], and plant-level optimization [12], [13]. Explainable-AI approaches have also been applied to RO performance prediction [14], alongside long short-term memory networks [15] and Fourier neural operators [16]. Shahouni et al. [17] survey this literature and identify interpretability and reduced data requirements as open needs, a gap that physics-informed approaches are intended to close.

Physics-informed neural networks (PINNs) [18], [19] combine the expressive capacity of a neural network with a governing-equation-based loss term, and have recently been applied to closed-circuit RO [20] and fast 3D transport simulation for RO design [21]. Building on this line of work, Li and Li [1] proposed a PINN framework for a commercial spiral-wound RO element (FilmTec SW30-2540), using a composite loss $\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda \cdot \mathcal{L}_{physics}$, where the physics term is based on the solution-diffusion-with-defects transport model [22]–[24] and a power-law pressure-

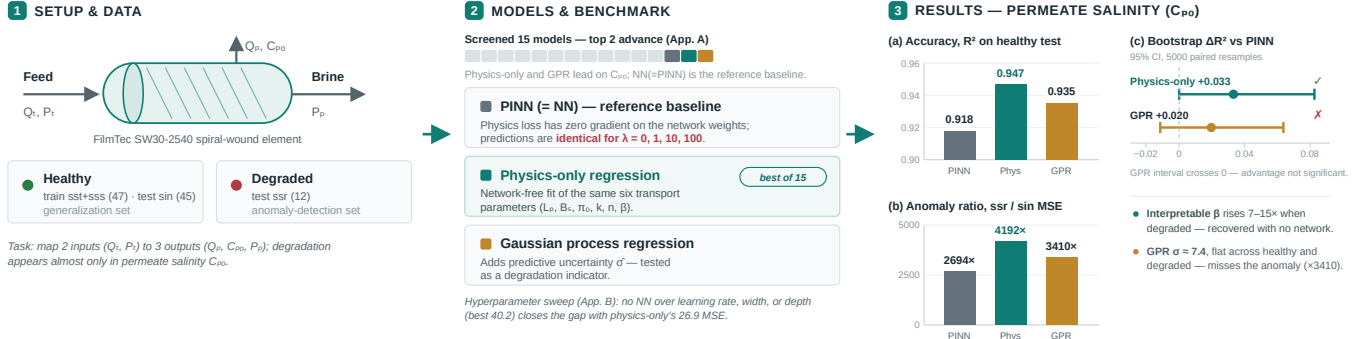


Fig. 1. Graphical abstract. On identical data and protocol, a network-free physics-only regression and Gaussian process regression are benchmarked against the reproduced PINN(=NN): physics-only is the best of the 15 screened models on permeate salinity (C_{po}), with a modest bootstrap-supported accuracy gain and a stronger anomaly signal, while the PINN’s contribution reduces to its interpretable output.

drop correlation. At $\lambda = 10$, the framework recovers six physically interpretable parameters: water permeability (L_p), salt permeability (B_s), feed osmotic pressure (π_0), a pressure-drop coefficient (k), a power-law index (n), and a defect fraction (β). The authors report that both the $\lambda = 0$ (data-only) and $\lambda = 10$ (physics-informed) configurations flag a degraded operating condition (the “ssr” dataset) via a large rise in prediction residual, with the $\lambda = 10$ configuration additionally attributing this degradation to a roughly order-of-magnitude increase in β .

That study’s own comparison, however, is confined to one architecture at two values of λ : what is described as “physics-informed neural network vs. data-driven model” is, architecturally, a single two-hidden-layer network evaluated with and without an auxiliary loss term. No comparison is made against classical regression, ensemble methods, or a physics-based model that omits the network entirely, so it remains unclear whether the network is contributing anything beyond what the governing equations alone would provide, and whether simpler, off-the-shelf regressors would perform comparably on both the accuracy and anomaly-detection tasks the paper reports.

This work addresses that gap directly, reusing the same public dataset [25], the same train/test protocol, and the same evaluation quantities as [1]. We make three contributions. First, we identify and resolve an ambiguity in the composite loss function’s physics term: the paper’s Eq. (7)-style normalized-mean-squared-error can be read as comparing the physics simulation against the network’s own output, or against the measured data directly; we show that only the latter reading reproduces the original paper’s reported Table 1 (in which the $\lambda = 0$ and $\lambda = 10$ configurations are nearly identical), and that under this reading the physics loss has strictly zero gradient with respect to the network weights, making the two configurations’ network component provably identical. Second, we benchmark the (now correctly interpreted) PINN against thirteen additional regressors, including a direct, network-free regression of the same six transport parameters (“physics-only”) and

Gaussian process regression (GPR), on both prediction accuracy and degradation-anomaly detection. Third, we test whether GPR’s predictive uncertainty, an output no other benchmarked model provides, can itself serve as a degradation indicator, and show that it cannot, because it reflects proximity to training inputs rather than correctness of the learned input-output mapping.

II. METHODS

Fig. 2 summarizes the overall methodology: the public dataset is split into training (sst+sss) and test sets (sin healthy, ssr degraded) exactly as in [1]; three model families are fit on the training set; all are evaluated on the same accuracy, anomaly-detection, and interpretability metrics; and the two headline comparisons (physics-only and GPR vs. NN(=PINN)) are additionally checked for robustness via bootstrap resampling and a hyperparameter sweep before drawing conclusions.

A. Dataset

We use the same publicly available dataset as [1], originally collected by Das et al. [25] on a single FilmTec SW30-2540 spiral-wound RO element under steady and periodically varying feed conditions, and organized into the same four categories: steady-state test (sst), sinusoidal feed test (sin), steady-state integrity check following sinusoidal testing (sss), and steady-state integrity check following rectified-sinusoidal testing (ssr). Input features are feed flow rate (Q_f) and feed pressure (P_f); targets are permeate flow rate (Q_p), permeate salt concentration (C_{po}), and brine pressure (P_b). As in the original study, models are trained on the pooled healthy datasets (sst + sss, 47 points total) and evaluated on the held-out healthy set (sin, 45 points, generalization test) and the degraded set (ssr, 12 points, anomaly-detection test). Rectified-sinusoidal operation has been shown to accelerate membrane degradation and increase salt passage relative to sinusoidal operation, which behaves similarly to steady state under equivalent average conditions [26].

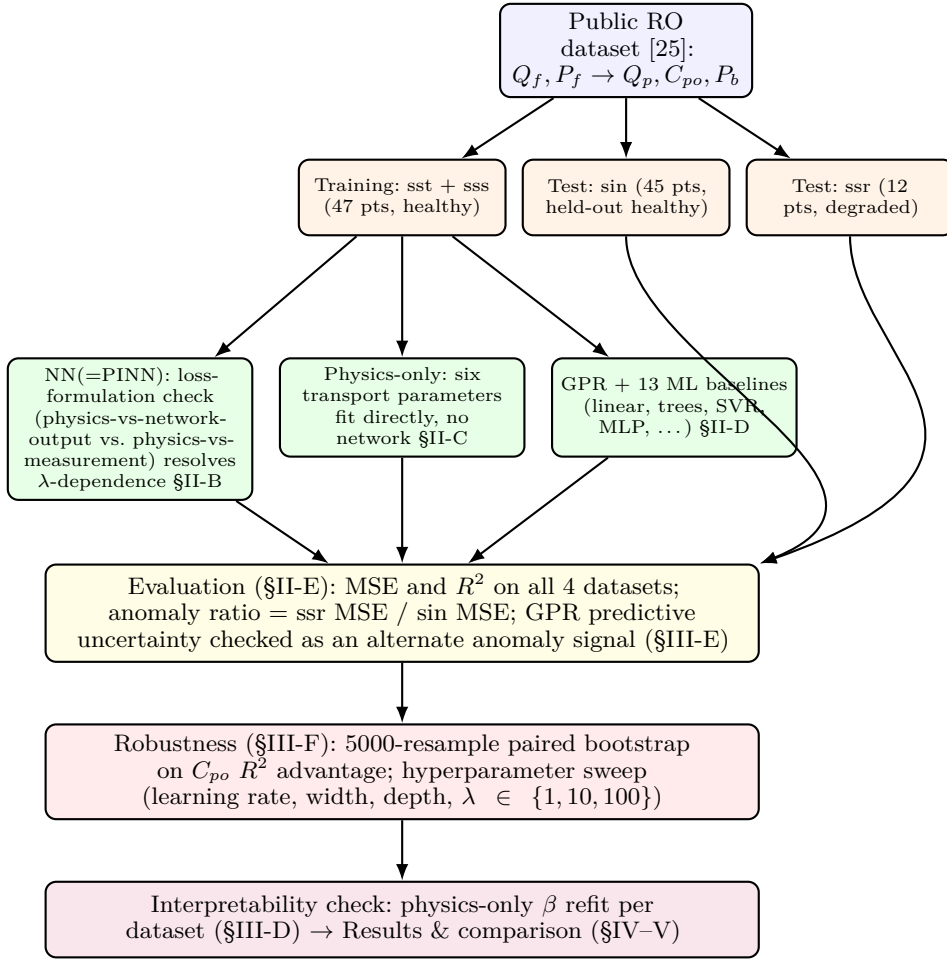


Fig. 2. Overview of the benchmarking methodology. All model families are trained on the identical healthy pool (sst+sss) and evaluated on the identical held-out sets, so accuracy, anomaly-detection, and interpretability comparisons are apples-to-apples across NN(=PINN), physics-only, and GPR.

B. Physics-Informed Neural Network: Reproduction and a Loss-Formulation Ambiguity

We reimplemented the network architecture and loss function of [1]: a two-hidden-layer, 20-neuron-per-layer feedforward network with tanh activations, mapminmax-style $[0, 1]$ input/output normalization, Adam optimization (learning rate 10^{-3}), mini-batches of size 32, and up to 5000 training epochs, with inputs shuffled each epoch. Writing the normalized input as $x = (Q_f, P_f)$ and the normalized output as $\hat{y} = (\hat{Q}_p, \hat{C}_{po}, \hat{P}_b)$, the network computes

$$\hat{y} = W_3 \tanh(W_2 \tanh(W_1 x + b_1) + b_2) + b_3, \quad (1)$$

with weight matrices W_1, W_2, W_3 and bias vectors b_1, b_2, b_3 learned by Adam.

The physics term is based on the solution-diffusion-with-defects transport model, consistent with the convection-augmented solution-diffusion framework described by Li and Li [27] and integrated along the membrane area. For feed flow Q_f and feed pressure P_f , with retentate pressure P_r , permeate pressure P_p , wall and permeate

osmotic pressures π_w, π_p , and bulk/wall/permeate salt concentrations C, C_w, C_p , the local water flux J_w and salt flux J_s are

$$J_w = L_p [(P_r - P_p) - (\pi_w - \pi_p)] + \beta L_p (P_r - P_p), \quad (2)$$

$$J_s = B_s (C_w - C_p) + \beta L_p (P_r - P_p) C, \quad (3)$$

where β is the fraction of membrane area affected by defects. Integrating over the membrane area A_m gives the module permeate concentration

$$C_{po} = \frac{\int_0^{A_m} J_s dA_m}{\int_0^{A_m} J_w dA_m}, \quad (4)$$

and the retentate pressure drop follows a power law in feed and recovery-adjusted flow Y ,

$$\Delta P_L = k \left[\frac{Q_f + Q_f(1 - Y)}{2} \right]^n, \quad (5)$$

with six learnable transport parameters $(L_p, B_s, \pi_0, k, n, \beta)$, where π_0 sets the feed osmotic pressure scale used in π_w, π_p . These four equations, following [1], are optimized

jointly with the network weights (parameter learning rate multiplied $\times 10$ relative to the network, following [1]).

The composite loss is

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda \cdot \mathcal{L}_{physics}, \quad (6)$$

where $\mathcal{L}_{physics}$ sums the normalized mean squared error (Eq. (7) of [1]) between physics-simulated and reference values of C_{po} , water flux J_w , and pressure drop ΔP_L . The reference paper’s Eq. (7) states this as a comparison between “measured and physics-based values,” but does not specify whether, during network training, the “measured” role is played by the raw experimental measurement or by the network’s own current prediction (a standard PINN residual formulation). We implemented both readings as a configurable option and trained under each:

- 1) Physics-against-network-output: the physics simulation is compared against the network’s own prediction at each training step, so $\mathcal{L}_{physics}$ depends on the network weights through the network’s output, and gradients from the physics loss reach the network.
- 2) Physics-against-measurement (literal reading of Eq. (7)): the physics simulation is compared directly against the experimental measurement, independent of the network’s current output.

Under reading (1), our reproduction’s PINN ($\lambda = 10$) sin-set C_{po} MSE was 1035.8 (mean of 5 seeds), with a standard deviation of over 1500 across seeds, an order of magnitude worse and far less stable than the $\lambda = 0$ network’s 52.4. Under reading (2), the PINN’s sin-set C_{po} MSE was 42.2 ± 10.0 , closely matching the original paper’s reported 44 ± 5 [1, Table 1], as did every other dataset/output combination in Table I (Section III-B). We therefore adopt reading (2) throughout, consistent with the original paper’s literal Eq. (7) and its reported near-identical NN/PINN predictions.

Under reading (2), $\mathcal{L}_{physics}$ is a function of the six transport parameters alone and does not depend on the network weights, so its gradient with respect to the network weights is exactly zero for any λ . The network weights are therefore updated only by $\mathcal{L}_{data}$, independent of λ , and a network trained at $\lambda = 10$ is (to floating-point precision, given the same random seed and no other source of stochasticity) identical to one trained at $\lambda = 0$. We verified this directly: across five seeds and all three outputs, the $\lambda = 0$ and $\lambda = 10$ networks’ predictions agreed to full double-precision floating-point precision on every dataset. Under this formulation, training at $\lambda > 0$ adds a second, independently-optimized regression of the six transport parameters alongside the (unaffected) network, rather than regularizing the network with physics. We refer to this configuration simply as NN(=PINN) in what follows, since its predictive component is common to both λ settings; the $\lambda = 10$ configuration’s only additional output is the learned parameter set, discussed in Section III-E.

C. Physics-Only Regression (No Network)

As a network-free baseline, we fit the same six transport parameters directly to data by minimizing $\mathcal{L}_{physics}$ alone (Adam, 4000 steps, learning rate 10^{-2}), with π_0 and n fixed (25 bar, 1.5, respectively, following [1, Section 3.3]) and B_s constrained to ≥ 0.04 lmh. This is the same “physics-only cross-check” the original study uses to sanity-check its individually trained PINNs, but here it is evaluated as a standalone predictive model on the full train/test protocol described in Section II-A, not only as a parameter cross-check.

D. Gaussian Process Regression and Additional Benchmark Models

We benchmarked thirteen additional regressors under the identical train/test protocol: linear and ridge regression, second-degree polynomial regression, k-nearest neighbors, random forest, extra-trees, gradient boosting, AdaBoost, extreme gradient boosting (XGBoost), support vector regression (SVR), a multilayer perceptron, and Gaussian process regression (GPR) [28], [29]. For training inputs X , targets y , and a query point x_* , GPR places a zero-mean Gaussian process prior with kernel $k(\cdot, \cdot)$ over the regression function and predicts

$$\mu_*(x_*) = k_*^\top (K + \sigma_n^2 I)^{-1} y, \quad (7)$$

$$\sigma_*^2(x_*) = k(x_*, x_*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_*, \quad (8)$$

where $K_{ij} = k(x_i, x_j)$, $k_* = [k(x_*, x_i)]_i$, and σ_n^2 is the observation noise variance. We used a constant $\times$ radial-basis-function kernel plus a white-noise term,

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{1}{2} \sum_d \frac{(x_d - x'_d)^2}{\ell_d^2}\right) + \sigma_n^2 \delta_{x, x'}, \quad (9)$$

with per-dimension length scales ℓ_d fit by maximizing the marginal likelihood. All models used standardized inputs; multi-output regressors not natively supporting three simultaneous targets were wrapped with independent per-output estimators. Stochastic models (random forest, extra-trees, gradient boosting, AdaBoost, multilayer perceptron, XGBoost) were averaged over multiple random seeds; GPR and the physics-only fit converged deterministically given their fixed optimization settings.

GPR was of particular interest because, unlike every other benchmarked model, it provides a predictive standard deviation $\sigma_*(x_*)$ (Eq. 8) alongside each point estimate $\mu_*(x_*)$ (Eq. 7), in principle enabling anomaly detection via predictive uncertainty rather than (or in addition to) residual error. We extracted this per-output uncertainty directly from the fitted per-output GPR estimators and compared its behavior on the healthy and degraded test sets (Section III-F).

E. Evaluation Protocol

For each model, output, and dataset with N points, measurements y_i , predictions $\hat{y}_i$, and mean $\bar{y} = \frac{1}{N} \sum_i y_i$, we report three quantities. First, the mean squared error,

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (10)$$

on all four datasets, following the original study's Table 1 format. Second, the coefficient of determination,

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (11)$$

computed independently per dataset using that dataset's own mean as the baseline, which is scale-free and comparable across outputs of different physical units. Third, an anomaly ratio,

$$\text{AR} = \frac{\text{MSE}_{ssr}}{\text{MSE}_{sin}}, \quad (12)$$

the ratio of degraded-set (ssr) to held-out-healthy-set (sin) MSE, following the original study's use of rising residuals as a degradation indicator. Where a model's advantage over the NN(=PINN) baseline is reported, it is the difference

$$\Delta R^2 = R^2_{\text{model}} - R^2_{\text{NN(=PINN)}}, \quad (13)$$

evaluated on the sin dataset. For the physics-only model, we additionally refit the six transport parameters separately on each of the four datasets (rather than only on the pooled sst+sss training set) to recover the degradation-sensitive parameter β , reproducing the original study's Table 2 protocol.

III. RESULTS AND DISCUSSION

A. Initial Screening Across Fifteen Models

Before the detailed comparison in the remainder of this section, all fifteen models (NN, PINN, physics-only, and the twelve additional regressors of Section II-D) were first screened together on the same accuracy (sin C_{po} MSE) and anomaly-detection (ssr/sin ratio) axes. Fig. 3 summarizes the result as a rank heat map across all three outputs and both axes, colour-coded within each column (green = best, red = worst; annotated values are the raw metrics, not the colour-mapped rank). Physics-only and GPR are the top two performers on the C_{po} columns, the axis that carries the real degradation signal (Section III-D), while the remaining eleven regressors (linear/ridge regression, polynomial regression, random forest, extra-trees, gradient boosting, AdaBoost, XGBoost, SVR, MLP, and k-nearest neighbors) cluster behind them on both accuracy and detection strength. We therefore carry physics-only and GPR forward for the full accuracy, anomaly-detection, interpretability, and robustness analysis in the remainder of this section; per-model results for all fifteen models are tabulated in Table VI (Appendix A).

B. The Physics Loss Formulation Determines Whether "PINN" Differs From "NN"

Table I reports MSE (mean $\pm$ standard deviation over 5 training runs) for NN and PINN under the literal (physics-against-measurement) loss reading, alongside the physics-only regression and GPR (single deterministic fit; standard deviation is exactly zero and is omitted for brevity where uninformative).

These values closely reproduce the original study's own Table 1 (e.g., sin C_{po} : 44 ± 5 reported vs. 42.2 ± 10.0 here; sst: 25 ± 2 to 26 ± 7 reported vs. 24.9 ± 3.9 here; ssr: $\sim 1.1 \times 10^5$ in both), confirming that the physics-against-measurement reading is the one consistent with the published results.

Fig. 4 shows measured-vs-predicted parity plots for physics-only and GPR across all three outputs and all four datasets, in the same style as the original study's own Fig. 1. The degraded (ssr) points fall on the diagonal for Q_p and P_b but sit well below it for C_{po} in both models: both models systematically under-predict permeate salinity for the degraded membrane, which is the visual signature underlying the anomaly ratios in Table III.

C. Accuracy: R^2 on Held-Out Healthy Data

Table II reports R^2 on the sin dataset (held-out healthy operating conditions), the standard measure of generalization used throughout this study.

Q_p and P_b are already predicted with $R^2 \geq 0.95$ by every model and do not meaningfully differentiate them. C_{po} , the output through which membrane degradation manifests (Section III-D), is where the models separate: physics-only reduces unexplained variance by 34.8% relative to NN(=PINN) (from 0.082 to 0.053), and GPR by 20.4% (to 0.065). Section III-G tests whether these point-estimate gaps hold up under bootstrap resampling of the test set; the physics-only gap does, the GPR gap does not.

To make Eq. 11 concrete for the best result in Table II, the sin dataset has $N = 45$ points with mean permeate salinity $\bar{y} = 132.11$ ppm, giving $SS_{tot} = \sum_i (y_i - \bar{y})^2 = 22,691.2$ ppm². The physics-only model's sin C_{po} MSE is 26.9 (Table I), so $SS_{res} = N \cdot \text{MSE} = 45 \times 26.9 = 1211.9$ ppm². Substituting into Eq. 11,

$$R^2_{\text{physics-only}} = 1 - \frac{1211.9}{22,691.2} = 1 - 0.0534 = 0.9466 \approx 0.947, \quad (14)$$

the value reported in Table II. The corresponding advantage over NN(=PINN) ($R^2 = 0.918$, $SS_{res} = 45 \times 42.2 = 1899.0$) is, by Eq. 13, $\Delta R^2 = 0.9466 - 0.918 = +0.033$, the same figure reported as the bootstrap point estimate in Table V (Section III-G) and carried through the Discussion and Conclusion below.

D. Anomaly Detection

Table III reports the anomaly ratio (ssr MSE / sin MSE) for C_{po} , the quantity the original study uses to demonstrate anomaly-detection capability.

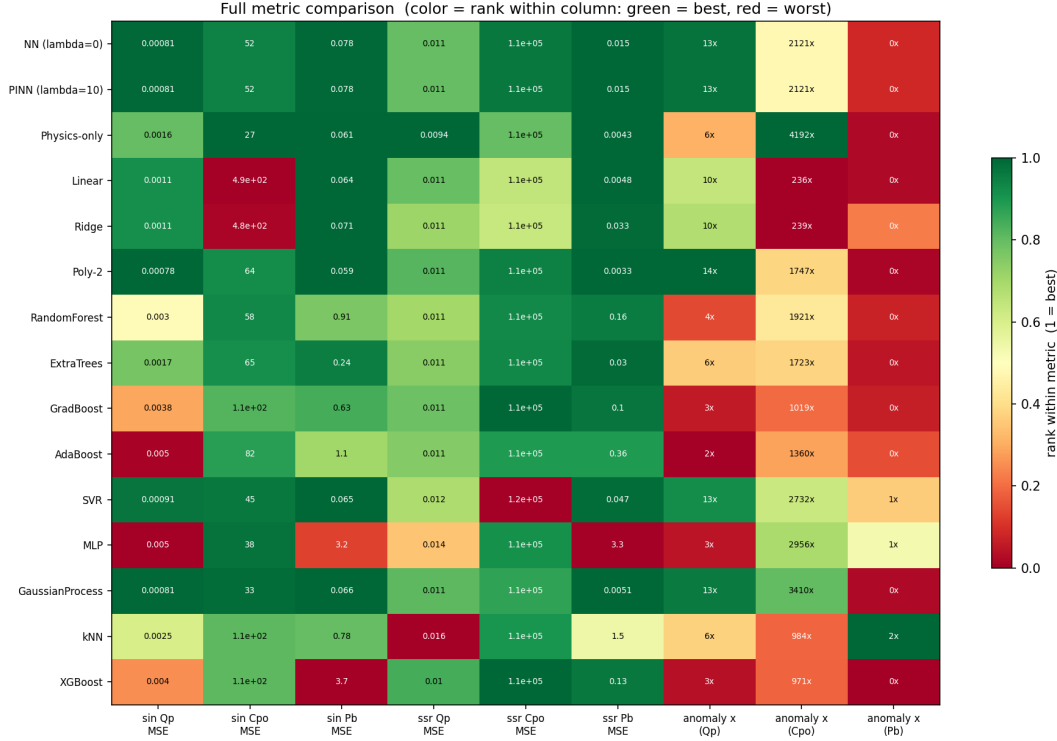


Fig. 3. Screening heat map across all 15 candidate models (rows) and 9 metrics (columns: MSE on sin/ssr and anomaly ratio, for Q_p , C_{po} , P_b). Colour indicates rank within each column (green = best, red = worst). Physics-only and GPR are consistently near the top of the C_{po} columns, motivating their selection for the detailed comparison in Sections III-B–III-G.

TABLE I
MEAN SQUARED ERROR (MSE) BY DATASET AND MODEL. NN AND PINN VALUES ARE REPRODUCED UNDER THE PHYSICS-AGAINST-MEASUREMENT LOSS READING (SECTION II-B) AND ARE NUMERICALLY IDENTICAL TO FULL FLOATING-POINT PRECISION; UNITS ARE L/MIN, PPM, AND BAR FOR Q_p , C_{po} , AND P_b , RESPECTIVELY.

Dataset	NN=PINN			Physics-only			GPR		
	Q_p	C_{po}	P_b	Q_p	C_{po}	P_b	Q_p	C_{po}	P_b
sst	$(3.83 \pm 0.16) \times 10^{-4}$	24.9 ± 3.9	$(1.11 \pm 0.28) \times 10^{-2}$	2.14×10^{-3}	65.4	5.25×10^{-3}	3.99×10^{-4}	33.9	6.65×10^{-3}
sss	$(9.11 \pm 0.52) \times 10^{-4}$	66.9 ± 6.2	$(3.29 \pm 0.93) \times 10^{-2}$	1.71×10^{-3}	62.1	2.27×10^{-2}	8.62×10^{-4}	66.6	2.19×10^{-2}
sin	$(7.28 \pm 0.40) \times 10^{-4}$	42.2 ± 10.0	$(6.51 \pm 1.08) \times 10^{-2}$	1.64×10^{-3}	26.9	6.07×10^{-2}	8.10×10^{-4}	32.9	6.58×10^{-2}
ssr	$(1.04 \pm 0.04) \times 10^{-2}$	$(1.11 \pm 0.006) \times 10^5$	$(1.14 \pm 0.43) \times 10^{-2}$	9.40×10^{-3}	1.13×10^5	4.34×10^{-3}	1.06×10^{-2}	1.12×10^5	5.11×10^{-3}

TABLE II
COEFFICIENT OF DETERMINATION (R^2) ON THE HELD-OUT HEALTHY (SIN) DATASET.

Model	Q_p	C_{po}	P_b
NN(=PINN)	0.979	0.918	0.997
Physics-only	0.953	0.947	0.997
GPR	0.977	0.935	0.997

TABLE III
ANOMALY RATIO (DEGRADED/HEALTHY MSE) FOR PERMEATE SALT CONCENTRATION.

Model	Anomaly ratio
NN(=PINN)	2694×
Physics-only	4192× (1.56× stronger)
GPR	3410× (1.27× stronger)

Both alternative models detect the degradation more strongly than NN(=PINN), while Q_p and P_b anomaly ratios remain near unity for every model (Table I), confirming, consistent with the original study, that the degradation signature in this dataset is concentrated in permeate salinity rather than flow or pressure.

E. Interpretability Without a Network

Refitting the physics-only model separately on each dataset (Section II-E) recovers a defect fraction β of 6.75×10^{-4} (sst), 3.05×10^{-4} (sss), and 5.04×10^{-4} (sin) on the three healthy datasets, rising to 4.60×10^{-3} on the degraded (ssr) dataset, a 7–15× increase, with L_p , B_s , and k essentially unchanged (Table IV). This reproduces the original study’s central diagnostic finding (a β increase of

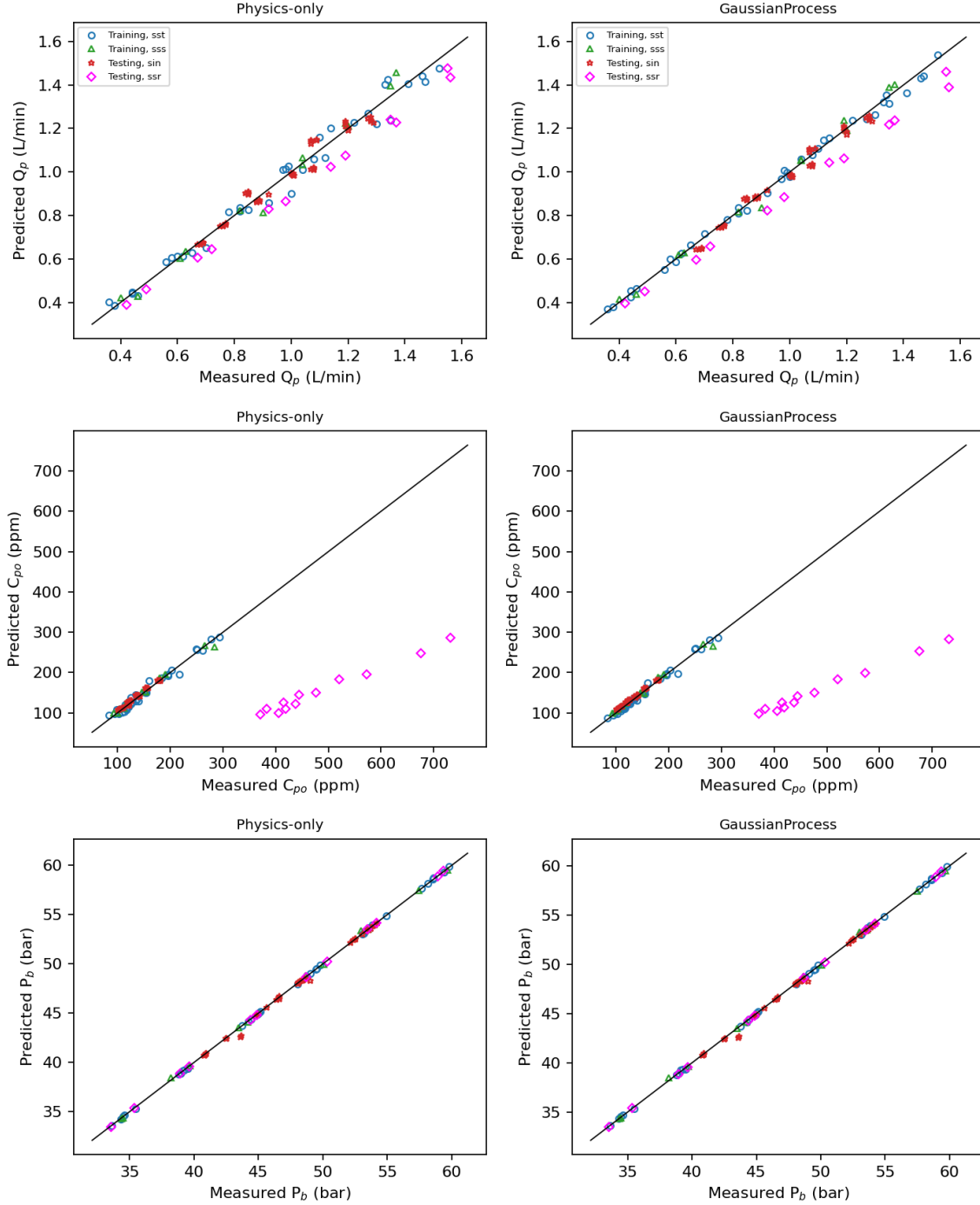


Fig. 4. Measured vs. predicted parity plots for Physics-only (left column) and GPR (right column), for Q_p (top), C_{po} (middle), and P_b (bottom), across all four datasets. The degraded (ssr) points depart from the diagonal only in the C_{po} panels, consistent with Table III.

similar order, attributed to structural loosening rather than fouling) without any network at all, and with substantially lower run-to-run variance than the individually-trained PINN, whose L_p standard deviation across seeds (0.26–0.39, our reproduction) is two orders of magnitude larger than the physics-only fit’s (exactly zero, given deterministic optimization from a fixed initial point).

F. Gaussian Process Uncertainty Does Not Detect the Anomaly

GPR’s predictive standard deviation for C_{po} was 7.63 (sst), 7.44 (sss), 7.41 (sin), and 7.48 (ssr), flat to within 1% across healthy and degraded conditions, despite a $3410\times$ rise in squared residual error over the same comparison.

Physics-only vs GaussianProcess — accuracy across all four datasets

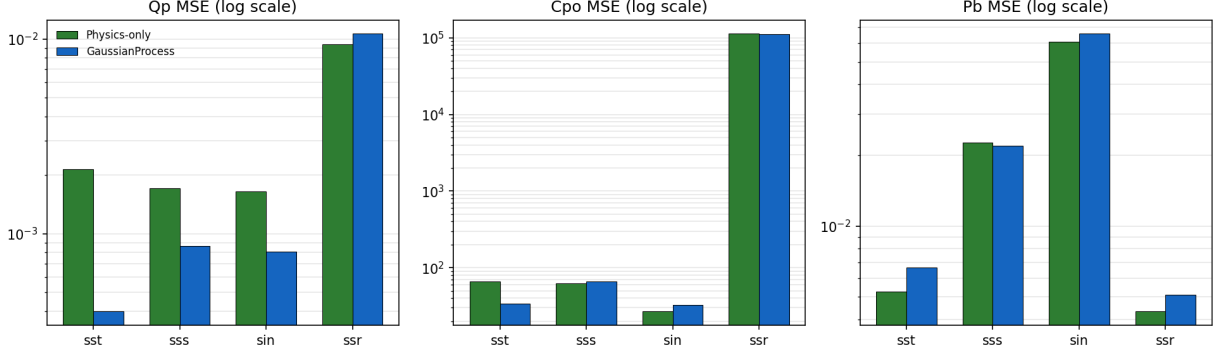


Fig. 5. MSE by output and dataset, Physics-only vs. GPR (log scale). C_{po} is the only output where a consistent, visible gap between models appears.

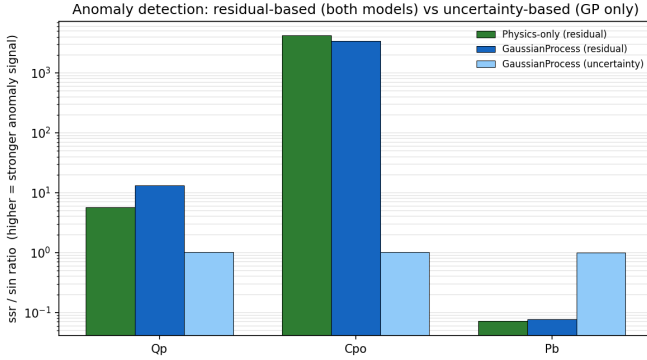


Fig. 6. Anomaly ratio for C_{po} : residual-based detection for Physics-only and GPR compared to NN(=PINN), alongside GPR’s uncertainty-based ratio (Section III-F), which is near $1\times$ and detects nothing.

TABLE IV
PHYSICS-ONLY TRANSPORT PARAMETERS, REFIT INDEPENDENTLY ON EACH DATASET.

Dataset	L_p (lmh/bar)	B_s (lmh)	k (bar/(L/min) ^{1.5})	β
sst	0.953	0.0454	0.0250	6.75×10^{-4}
sss	0.934	0.0585	0.0267	3.05×10^{-4}
sin	0.947	0.0500	0.0223	5.04×10^{-4}
ssr	1.024	0.0433	0.0247	4.60×10^{-3}

This is consistent with the standard behavior of stationary-kernel Gaussian process regression, whose predictive variance is a function of the query point’s proximity to training inputs in feature space, not of whether the fitted input-output mapping remains correct at that point, an instance of the broader distinction between epistemic (“knownness”) and aleatoric uncertainty [30], and a known failure mode of distance-based uncertainty estimates under covariate-preserving distribution shift [31]. Because the ssr dataset’s feed flow and pressure conditions fall within the same operating range as the healthy training data, GPR has

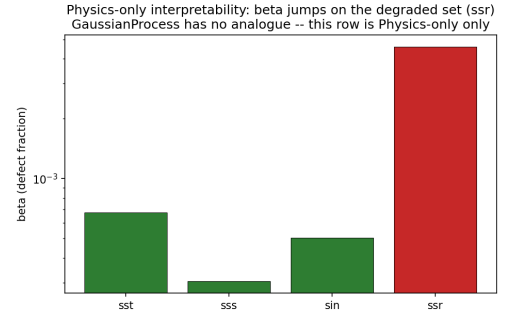


Fig. 7. Physics-only defect fraction β , refit independently on each dataset (log scale). β is stable across the three healthy datasets and rises 7–15 $\times$ on the degraded (ssr) dataset.

no basis in its input distribution to flag anything unusual, even though the true output relationship has shifted. This indicates that GPR’s “free” uncertainty, often cited as an advantage over point-estimate models [28], [29], is not a reliable substitute for residual monitoring in this class of degradation-diagnosis problem, where degradation manifests as a change in the input-output mapping rather than a shift in the operating envelope itself.

G. Robustness: Bootstrap Confidence Intervals and Hyperparameter Sensitivity

The point estimates in Sections III-B–III-C are computed on one fixed 45-point held-out set, so we tested whether they survive resampling. We drew 5000 paired bootstrap resamples of the sin test set, computing each replicate’s C_{po} R^2 for physics-only, GPR, and NN(=PINN) (with a randomly selected one of its five trained seeds, folding in training-seed variability alongside test-set sampling variability) on the same resampled points. Table V reports the resulting R^2 advantage over NN(=PINN).

Physics-only’s advantage is modest but the interval excludes (or, at its lower bound, just touches) zero, consistent with a real, if small, effect. GPR’s interval spans zero

TABLE V
BOOTSTRAP 95% CONFIDENCE INTERVALS FOR THE C_{po} R^2
ADVANTAGE OVER NN(=PINN) ON HELD-OUT HEALTHY DATA
(5000 RESAMPLES).

Model	Mean ΔR^2	95% CI	$P(\Delta R^2 \leq 0)$
Physics-only	+0.033	[0.000, 0.082]	0.023
GPR	+0.020	[-0.011, 0.063]	0.348

comfortably, and its advantage is not statistically distinguishable from noise given this test set’s size, a materially weaker claim than the point estimate in Section III-B alone would suggest, and one this study did not initially report before applying this check.

We additionally tested whether a more extensively tuned network could close the gap with physics-only, sweeping the same grid used in the original study’s own supplementary sensitivity analysis: learning rate $\in \{10^{-2}, 10^{-3}, 10^{-4}\}$, network width $\in \{10, 20, 50\}$ neurons, and depth $\in \{1, 2, 3\}$ hidden layers (each averaged over 5 seeds, all other settings at their default values). The best configuration found (3 hidden layers) reached $\sin C_{po}$ MSE of 40.2 ± 6.5 , better than the default 2-layer network’s 42.2 ± 10.0 , but still well short of physics-only’s 26.9. We further confirmed, training explicitly at $\lambda \in \{1, 10, 100\}$ under the physics-against-measurement reading, that predictions are bit-for-bit identical across this entire range and to $\lambda = 0$, extending the result of Section II-B from a single $\lambda = 10$ check to the original study’s full tested range.

IV. DISCUSSION

Taken together, these results narrow rather than overturn the original study’s claims, and the bootstrap analysis (Section III-G) narrows them further than the point estimates alone. The PINN framework’s reported ability to detect degradation via rising residuals is confirmed and, under the literal reading of its own loss function, is mathematically inherited entirely from the underlying neural network: the physics term does not alter the network’s predictions at all, across its entire tested λ range. What the physics term adds is a second, interpretable output (the six transport parameters, most importantly β), obtained at the cost of a joint optimization that, in our reproduction, is less stable (larger run-to-run parameter variance) than fitting the same equations directly with no network. A direct physics-only regression provides the same interpretability with a modest, bootstrap-supported accuracy edge, a stronger anomaly signal, and no training instability, at a fraction of the implementation complexity, and this edge persists even after a hyperparameter sweep over the network’s learning rate, width, and depth.

GPR’s case is weaker than physics-only’s. Its point-estimate accuracy and anomaly-ratio improvements over NN(=PINN) are comparable in size to physics-only’s, but the bootstrap interval on its accuracy advantage spans zero: on this dataset, we cannot rule out that GPR’s

apparent edge is sampling noise from a 45-point test set. We therefore do not claim GPR reliably outperforms the plain network here, only that its point estimate does not underperform it either, while requiring no physics assumptions. For practitioners working with a membrane whose governing equations are not well characterized, this makes GPR a reasonable, low-assumption alternative to try (its predictive-uncertainty output is a genuinely useful property in general), but Section III-F shows that uncertainty should not be relied upon as a degradation indicator in place of residual monitoring for this class of problem.

These findings should be read against several remaining limitations. First, both the point estimates and the bootstrap analysis are based on a small dataset (47 training points, 45 held-out healthy, 12 degraded), and the bootstrap resamples the existing test set rather than generating new data, so it quantifies sensitivity to which of these particular 45 points are evaluated, not to the underlying measurement process more broadly. Second, our physics-only and GPR models used a single, reasonable configuration for parameters not swept in Section III-G (e.g., the GPR kernel form, the physics-only optimizer’s step count); a wider search could shift either model’s result somewhat. Third, the loss-formulation ambiguity identified in Section II-B is a property of how Eq. (7) of the original study is implemented, not evidence of an error in the original results, which our reproduction (Section III-B) closely matches under the literal reading; we raise it because the same ambiguity could lead future implementations of similar composite-loss frameworks to unintentionally couple or decouple the physics term from the network, with materially different outcomes, as shown directly by the order-of-magnitude accuracy and stability difference between the two readings we tested.

V. CONCLUSION

We benchmarked a published PINN framework for reverse osmosis membrane performance modeling and degradation diagnosis against thirteen classical and ensemble machine-learning models and a network-free physics-only regression, using the same public dataset and evaluation protocol as the original study. We show that, under a literal reading of the study’s own composite loss function, the physics term is mathematically decoupled from the network weights across its entire tested λ range (1, 10, 100), so the reported PINN and its purely data-driven counterpart share an identical predictive model; this reading, and not the alternative (network-coupled) one, is what reproduces the original study’s published results, and no swept learning rate, network width, or depth closes the resulting accuracy gap. Against this corrected baseline, a direct regression of the same six transport parameters, with no network at all, achieves a modest, bootstrap-supported accuracy advantage ($\Delta R^2 = +0.033$, 95% CI [0.000, 0.082]) and a 56% stronger degradation-anomaly

TABLE VI

FULL SCREENING RESULTS, ALL 15 MODELS (SINGLE RUN AT DEFAULT SETTINGS; TABLE I REPORTS THE SEED-AVERAGED VALUES FOR THE TWO FINALISTS).

Model	$\sin Q_p$ MSE	$\sin C_{po}$ MSE	$\sin P_b$ MSE	ssr C_{po} MSE	Anomaly ratio (C_{po})
NN($\lambda = 0$) / PINN($\lambda = 10$)	8.07×10^{-4}	52.35	7.83×10^{-2}	111057.0	2121
Physics-only	1.64×10^{-3}	26.93	6.07×10^{-2}	112874.5	4192
Linear regression	1.13×10^{-3}	485.55	6.44×10^{-2}	114610.3	236
Ridge regression	1.14×10^{-3}	479.74	7.08×10^{-2}	114720.3	239
Polynomial (deg. 2)	7.80×10^{-4}	63.68	5.91×10^{-2}	111223.2	1747
Random forest	2.97×10^{-3}	58.01	9.12×10^{-1}	111422.0	1921
Extra-trees	1.75×10^{-3}	64.65	2.39×10^{-1}	111408.9	1723
Gradient boosting	3.80×10^{-3}	108.55	6.33×10^{-1}	110639.7	1019
AdaBoost	5.00×10^{-3}	82.25	1.12	111830.8	1360
Support vector (RBF)	9.09×10^{-4}	44.60	6.48×10^{-2}	121836.4	2732
Multilayer perceptron	5.04×10^{-3}	37.76	3.18	111608.5	2956
Gaussian process (GPR)	8.10×10^{-4}	32.88	6.58×10^{-2}	112117.8	3410
k-nearest neighbors	2.47×10^{-3}	113.53	7.82×10^{-1}	111687.4	984
XGBoost	3.96×10^{-3}	113.89	3.66	110632.5	971

signal, while independently reproducing the study’s central interpretability result, a 7- to 15-fold rise in defect fraction β on degraded membrane data, with lower run-to-run variance. Gaussian process regression shows a comparable point-estimate improvement (20% lower error, 27% stronger anomaly signal) without assuming any transport equations, but this advantage does not survive bootstrap resampling of the test set (95% CI $[-0.011, 0.063]$) and should not be treated as established; its predictive uncertainty is, in any case, shown to be uninformative for this anomaly-detection task, since it reflects distance from training inputs rather than correctness of the learned mapping. For this dataset, the physics-informed network’s value lies in its interpretable output rather than in a strongly established predictive advantage over simpler alternatives, a distinction that a broader, statistically-checked benchmark than the original study’s own two-configuration comparison was needed to reveal.

APPENDIX

Table VI reports the complete initial-screening results referenced in Section III-A for all 15 candidate models. NN and PINN are numerically identical (Section II-B), so they share one row.

Table VII lists the exact configuration used for each model. Stochastic models were run with the seed sets described in Section II-D; all others are deterministic given these settings.

Table VIII reports the full grid underlying the hyperparameter-sensitivity claim in Section III-G: no swept configuration of the NN(=PINN) closes the gap with physics-only’s $\sin C_{po}$ MSE of 26.9.

TABLE VII
MODEL CONFIGURATIONS.

Model	Configuration
NN(=PINN)	2 hidden layers, 20 neurons, tanh, Adam (10^{-3}), batch 32, 5000 epochs
Physics-only	Adam (10^{-2}), 4000 steps; $\pi_0 = 25$ bar, $n = 1.5$ fixed; $B_s \geq 0.04$ lmh
Linear / Ridge	OLS / $\alpha = 1.0$, standardized inputs
Polynomial (deg. 2)	Degree-2 features + linear regression
Random forest / Extra-trees	300 trees, default depth
Gradient boosting	200 estimators, default depth
AdaBoost	100 estimators
SVR	RBF kernel, $C = 10$, $\epsilon = 0.01$
MLP	2 hidden layers (20, 20), $\alpha = 10^{-4}$, max. 5000 iterations
GPR	Constant $\times$ RBF + white-noise kernel, length scales fit by marginal-likelihood maximization
k-NN	$k = 5$ neighbors
XGBoost	200 estimators, max. depth 3, learning rate 0.1

TABLE VIII
NN(=PINN) HYPERPARAMETER SWEEP, $\sin C_{po}$ MSE (MEAN $\pm$ SD, 5 SEEDS), ALL OTHER SETTINGS AT DEFAULT (LEARNING RATE 10^{-3} , 20 NEURONS, 2 LAYERS).

Swept setting	$\sin C_{po}$ MSE
Learning rate 10^{-2}	53.3 ± 29.9
Learning rate 10^{-3} (default)	42.2 ± 10.0
Learning rate 10^{-4}	41.5 ± 3.7
10 neurons	47.1 ± 3.7
20 neurons (default)	42.2 ± 10.0
50 neurons	41.7 ± 7.6
1 hidden layer	46.8 ± 7.4
2 hidden layers (default)	42.2 ± 10.0
3 hidden layers (best found)	40.2 ± 6.5
$\lambda \in \{0, 1, 10, 100\}$	42.231 ± 9.976 (bit-identical at every λ)

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STATEMENT ON GENERATIVE AI AND AI-ENABLED TECHNOLOGIES

The authors used OpenAI's ChatGPT and Anthropic's Claude in different aspects of the preparation of this work, including programming, language polishing, document organization, and preparation of supporting visual and mathematical material. Subsequently, the authors reviewed and validated all content, code, calculations, interpretations, and references generated with the assistance of AI. The authors assume full responsibility for the accuracy, originality, and integrity of the final manuscript.

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